



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

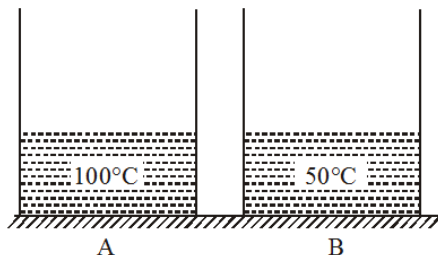
1. An electromagnetic wave of intensity 200 W/m^2 is incident on a rectangular black card with sides of 20 and 30 cm. Find the force exerted on the card by the radiation
 - (A) If the card reflects all the radiation and the radiation is incident normally, the force is $0.08 \mu\text{N}$
 - (B) If the card reflects all the radiation and the radiation is incident normally, the force is $0.06 \mu\text{N}$
 - (C) If the card absorbs all the radiation and the radiation is incident at an angle of 30° to the normal, the force is $2\sqrt{3} \times 10^{-8} \text{ N}$
 - (D) If the card reflects all the radiation and the radiation is incident at an angle of 30° to the normal, the force is $0.06 \mu\text{N}$

Space for Rough Work

2. A double-slit experiment uses a helium-neon laser with a wavelength of 600 nm and a slit separation of 0.12 mm. When a thin sheet of plastic is placed in front of one of the slits, the interference pattern shifts by 5.5 fringes. When the experiment is repeated under water $\left(\mu = \frac{4}{3}\right)$, the shift is 3.5 fringes.
- (A) The thickness of the plastic sheet is $3.6\mu\text{m}^{-1}$
- (B) The index of refraction of the plastic sheet is $\frac{6}{5}$
- (C) The thickness of plastic sheet is $4.2\mu\text{m}$
- (D) The index of refraction of plastic sheet is $\frac{23}{12}$
3. A metal wire of length L , area of cross section A and Young modulus Y is stretched by a variable force F such that F is always slightly greater than the elastic forces of resistance in the wire. When the elongation of the wire is ℓ , then choose the correct option (s).
- (A) The work done by F is $\frac{1}{2} \left(\frac{Y A}{L} \right) \ell^2$
- (B) The elastic potential energy stored in the wire is $\frac{1}{2} \left(\frac{Y A}{L} \right) \ell^2$
- (C) No heat is produced during the elongation
- (D) The work done by F is $\left(\frac{Y A}{L} \right) \ell^2$

Space for Rough Work

4. 2 kg water at 100°C and 2.5 kg water at 50°C is kept in two identical containers A and B respectively of water equivalent 0.5 kg. If water of container A is poured into container B the final temperature of mixture is T_1 and if water of container B is poured into container A the final temperature is T_2 . (heat loss is negligible)



- (A) $15T_1 = 14T_2$ (B) $T_1 = T_2$ (C) $T_1 = 70^{\circ}\text{C}$ (D) $T_2 = 70^{\circ}\text{C}$
5. The energy radiated by a black body at 2300 K is found to have the maximum at a wavelength 1260 nm, its emissive power being 8000 Wm^{-2} . When the body is cooled to a temperature T K, the emissive power is found to decrease to 500 Wm^{-2} .
- (A) $T = 1150 \text{ K}$
- (B) The wave length at which intensity of emission is maximum at the temperature $T = 2520 \text{ nm}$
- (C) $T = 4600 \text{ K}$
- (D) $\lambda_m = 630 \text{ nm}$
6. A rod of mass 1kg and length 3m is hinged at its end and can oscillate in a vertical plane about a horizontal axis. A periodic torque of $\tau = 0.5 \sin(2t) \text{ Nm}$ acts on the rod. (t is in sec.)
- (A) The angular amplitude of oscillation of the rod is $\frac{1}{6} \text{ rad}$.
- (B) The angular amplitude of oscillation of the rod is $\frac{1}{4} \text{ rad}$
- (C) The oscillation of the rod is in phase with the applied torque
- (D) The oscillation of the rod is out of phase with the applied torque

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

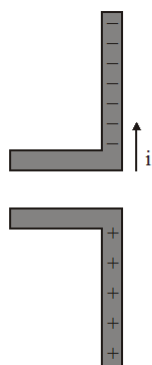
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

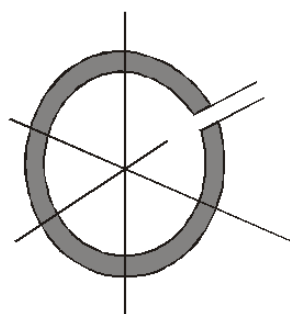
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

An electric-dipole radio antenna consists of two conducting rods along a line fed by an alternating-current generator. Similarly, we have a magnetic dipole antenna which is a conducting loop fed by an alternating current generator. Far from the antenna, the electric and magnetic fields oscillate in phase with simple harmonic motion, perpendicular to each other and to the direction of propagation of the wave.



Electric dipole antenna



magnetic dipole antenna

Electromagnetic waves of radio or television frequencies can be detected by a dipole antenna placed appropriately so that it induces an alternating current in the antenna. They can also be detected by a magnetic dipole antenna placed appropriately so that the changing magnetic flux through the loop induces a current in the loop.

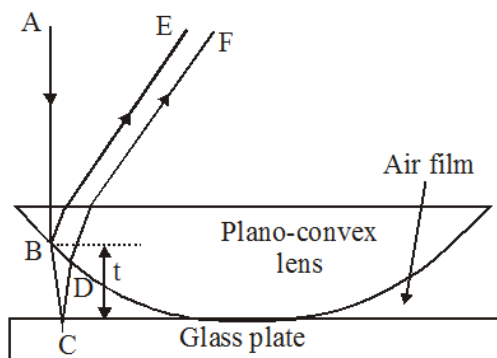
7. A transmitter uses a magnetic dipole antenna with the loop in the horizontal plane. What should be the orientation of an electric dipole antenna at the receiver for optimum signal reception? Assume that the position vector of the receiver with respect to the transmitter is horizontal.
- (A) Vertical
- (B) Horizontal with orientation perpendicular to the direction of position vector mentioned.
- (C) Horizontal with orientation parallel to the direction of position vector mentioned.
- (D) At this position, the receiver cannot receive any signal for any orientation of the receiving antenna.

8. A transmitter uses an electric dipole antenna with the rods horizontal oriented in the east west direction. What should be the orientation of a magnetic dipole antenna at the receiver for optimum signal reception? Assume that the position vector of the receiver with respect to the transmitter is horizontal and receiver is in a direction east of the transmitter.
- (A) loop should be horizontal
- (B) loop should be vertical with axis along east west direction.
- (C) loop should be vertical with axis along North south direction.
- (D) At this position, the receiver cannot receive any signal for any orientation of the receiving antenna.

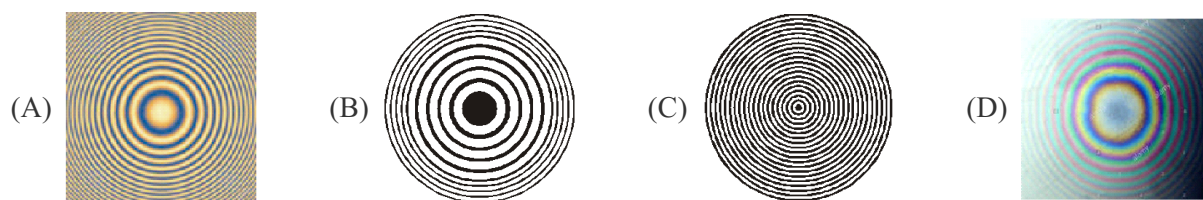
Space for Rough Work

Paragraph for Questions 9 and 10

In the original Newton's rings experiment shown here, we have an air film between a plano-convex glass lens kept over a glass plate as shown in the figure below. A modification to that experiment is done. We take a plano-convex lens of refractive index 1.3 and radius of curvature 100 cm. We keep it over a glass plate of refractive index 1.5 and fill a liquid of refractive index 1.4 in the space in between. We want to predict what will be the shape of interference fringes formed. We use monochromatic light of wavelength 630 nm which is incident normally from the top.



9. Which of the following correctly represents the shape of the interference fringes.



10. What is the radius of the 5th maxima?

- (A) 1.5 mm (B) $1.5\sqrt{\frac{11}{13}}$ mm (C) $1.5\sqrt{\frac{7}{13}}$ mm (D) $1.5\sqrt{\frac{11}{10}}$ mm

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

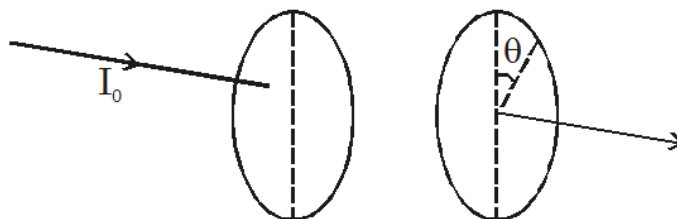
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. A bead of mass 6.63 pg (picogram) slides on a horizontal wire 2m long. It's speed is 10^{-19} m/s so that it is apparently at rest. Treat this system as a quantum particle in one dimensional box with infinitely high walls. Calculate the quantum number of state described.
 2. Atomic furnace wall has a hole having a diameter of δ . At a distance of L from the furnace wall, we kept a plane screen which can detect atoms striking it. Assume that atoms of mass m moving with velocity v in x direction emerge from the hole. Using de Broglie's hypothesis, find the smallest radius (in mm) of the spot on the screen. (Take : $h = 6.4 \times 10^{-34}$ kg m²/s, $L = 1$ m, $m = 1.6 \times 10^{-27}$ kg, $v = 10^3$ m/s, $\delta = 0.2$ μ m).
-

Space for Rough Work

3. Suppose that an unpolarized light beam is incident from the left on the arrangement of two polarizers illustrated in figure. If the intensity of the light emerging on the right is 30% of the incident intensity, what must be the angle θ between the preferential directions of the polarizers ? Fill $\cos \theta$



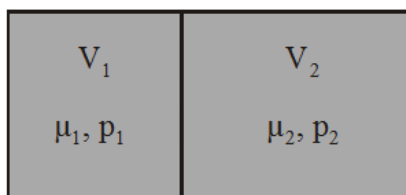
4. Determine the smallest angular separation (in 10^{-7} rad) between two equally bright stars that can be resolved (in the sense of the Rayleigh criterion) by the 5 m diameter objective Hale telescope on Mount Palomar. Take : $\lambda = 500$ nm

Space for Rough Work

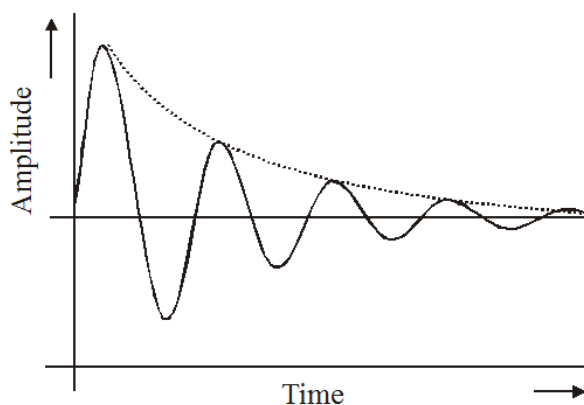
5. The prism angle of a very thin prism is measured by observing interference fringes as in the Fresnel biprism technique. The distances from slit to prism and from prism to screen are in the ratio of 1 : 4. Twenty dark fringes are found to span a distance of 0.5 cm when green mercury $\lambda = 490$ nm light is used. If the refractive index of the prism is 1.50, determine the prism angle (in m rad)
6. Call the intensity at the center of the central Fraunhofer diffraction maximum of a single slit I_0 and the intensity at some other point in the pattern I . Obtain the ratio I/I_0 for a point on the screen that is $3/4$ wavelength farther from one edge of the slit than the other.

Space for Rough Work

7. The adiabatic container shown in figure has two chambers, separated by a partition, of volumes $V_1 = 2.0$ litre and $V_2 = 3.0$ litre. The chambers contain $\mu_1 = 4.0$ and $\mu_2 = 5.0$ moles of a gas at pressures $p_1 = 1.00$ atm and $p_2 = 2.00$ atm. The pressure after the partition is removed and the mixture attains equilibrium is P_0 atm, then find the value of $2.5 P_0$.



8. A spring block system undergoes damped oscillation whose displacement time graph is shown below. The initial amplitude is 1 cm and after one oscillation, it becomes 0.37 cm. If the time period of oscillation is 4 sec, what is the damping constant b (in kg/sec)? Take the mass of the oscillator as 1 kg.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is(are) correct ?

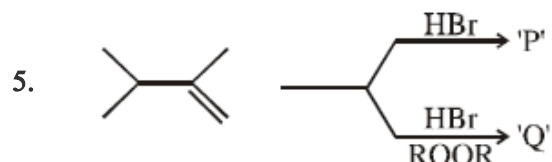
- (A) Number of geometrical isomers of ionisation isomers of $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{Cl}_2]\text{Br}$ is 5 (except the given compound)
- (B) In $\text{Fe}_2(\text{CO})_9$, $d_{\text{C-O}}$ (bridging) < $d_{\text{C-O}}$ (terminal)
- (C) IUPAC name of $[(\text{Cl}_3\text{Sn})_2\text{RhCl}_2\text{Rh}(\text{SnCl}_3)_2]^{4+}$ is Di- μ -chloridotetrakis(trichlorostannito)dirhodium(I) ion
- (D) Complexes $[\text{Co}(\text{NO}_2)_6]^{4-}$ and $[\text{CoF}_6]^{3-}$ are inner orbital complex and outer orbital complex respectively

Space for Rough Work

2. Which statement(s) can be possible for homonuclear diatomic neutral molecules formed by any 2nd period element.
- (A) Atleast one paramagnetic species has only σ bonds
 (B) Atleast one diamagnetic species has only σ bonds
 (C) Atleast one diamagnetic species has only π bonds
 (D) Atleast one paramagnetic species has only π bonds
3. EMF of the cell: $\text{Cd(s)} \mid \text{CdCl}_2 \cdot 5\text{H}_2\text{O (sat.)} \mid \text{AgCl(s)} \mid \text{Ag(s)}$ is +0.70 V at 0°C and +0.60 V at 50°C. If ΔH° and ΔS° are temperature independent, then the correct information(s) regarding the cell reaction is/are -
- (A) $\Delta G^\circ = -115.8 \text{ kJ at } 50^\circ\text{C}$ (B) $\Delta G^\circ = 135.1 \text{ kJ at } 0^\circ\text{C}$
 (C) $\Delta S^\circ = -386 \text{ J/K}$ (D) $\Delta H^\circ = -221 \text{ kJ}$

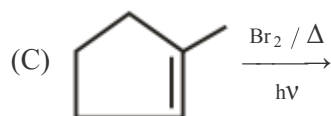
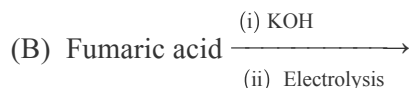
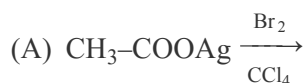
Space for Rough Work

4. 250 ml of 1M I_2 is separately reacted with 0.5M Cu_2S solution, 0.5M CuS solution and 0.5 M $S_2O_3^{2-}$ solution causing production of Cu^{+2} and SO_4^{2-} in the first two and $S_4O_6^{2-}$ in the last case along with iodide ions. Which of the following option is/are correct assuming 100% completion of the reaction. (I = 127)
- (A) 100 ml of Cu_2S solution will be consumed.
- (B) 125 ml of CuS solution will be consumed.
- (C) 500 ml of $S_2O_3^{2-}$ solution will be consumed.
- (D) Equivalent weight of I_2 in each of the reaction will be 127.



Correct statement regarding P & Q is/are

- (A) P & Q are structural isomer
- (B) P can show optical isomerism
- (C) Q can show optical isomerism
- (D) Q is tertiary bromide
6. In which reaction(s), free-radical intermediate is involved :



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

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Paragraph for Questions 7 and 8

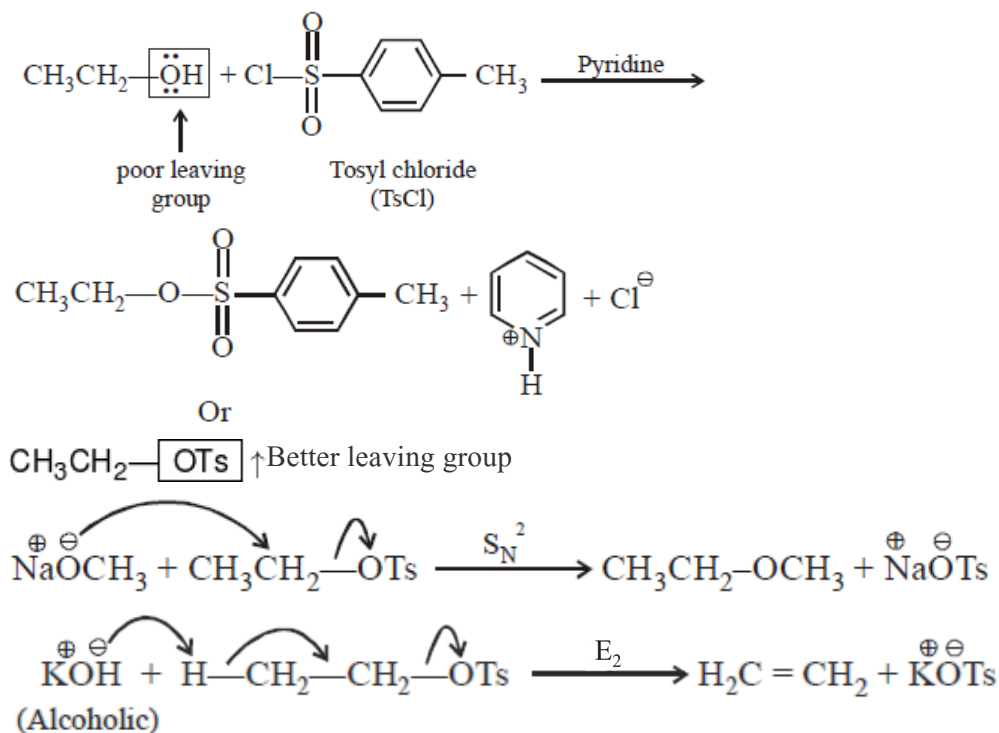
The complex HgCo_4SCN has two isomers X and Y (having Co as central metal ion in both) which are HOMOLEPTIC and magnetic moment (μ) of X = Y.

7. Choose the incorrect option for X and Y?
- (I) X and Y are different in shape
 (II) Oxidation of Hg and Co are same in both the complexes
 (III) X has spin only magnetic moment equals to 1.73 BM
 (IV) Both isomers having same hybridisation around central metal ion
- (A) I and II (B) III and IV (C) I and III (D) II, III and IV
8. The individual sum of the linkage isomers of 'X' and 'Y' are.
- (A) 8 (B) 6 (C) 10 (D) 4

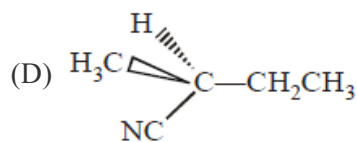
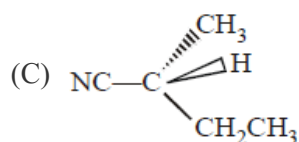
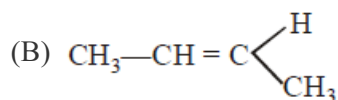
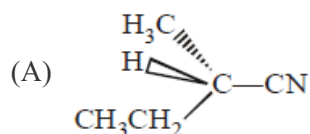
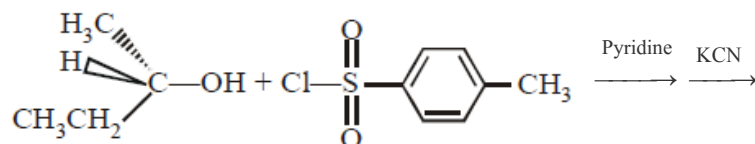
Space for Rough Work

Paragraph for Questions 9 and 10

Alcohols are converted to tosylates by treatment with p-toluene sulfonyl chloride (TsCl) in presence of pyridine. This overall process converts a poor leaving group (OH^-) into better one (OTs^-). A tosylate is a better leaving group because its conjugated acid p-toluene sulfonic acid is strong acid. Because alkyl tosylates have better leaving tendency they undergo both nucleophilic substitution and β -elimination.

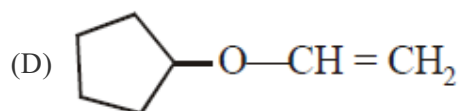
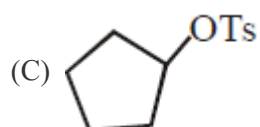
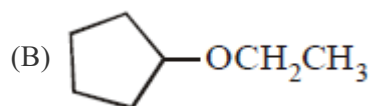
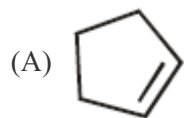
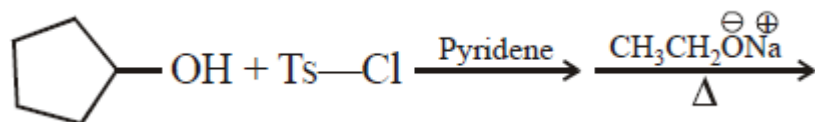


9. Find the major product of following reaction :



Space for Rough Work

10. What would be the major product of following reactions ?



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Find number of process or ore given in Column-I are correctly represented with Column-II. You have to consider P for P, Q for Q, R for R etc.

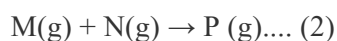
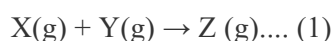
(I) Process/Ore		(II) Related to	
(P)	Matte smelting	(P)	Cu
(Q)	Cyanide leaching	(Q)	Au
(R)	Mond's process	(R)	Reaction of metal + H ₂
(S)	Fused oxide electrolysis	(S)	Al
(T)	Chromite ore	(T)	Fe + Cr
(U)	Van Arkel process	(U)	Zr
(V)	Bacterial leaching	(V)	Low grade Cu ore
(W)	Separation of FeWO ₄	(W)	Extraction of Pb

Space for Rough Work

2. Consider the following statements

- (1) $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightarrow 2\text{CO}(\text{g})$; it is an exothermic reaction
 - (2) Ilmenite is an carbonate ore containing two metals
 - (3) In froth floatation process KCN act as an activator when galena contains impurity of ZnS
 - (4) Liquation is applicable for the purification of Sn , Pb, Zn
 - (5) $\text{Au} + \text{NaCN} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{Na}[\text{Au}(\text{CN})_4]$ reaction occurs during leaching of gold
 - (6) Al_2O_3 can be obtained in solution from red bauxite by using concentrated NaOH
 - (7) MgO was reduced by C at very high temperature followed by a rapid cooling with the help of inert gas
- The number of correct statement(s) in the above given statements are :

3. For two reactions,



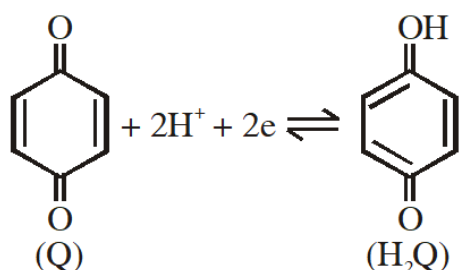
According to the collision theory, the ratio of squares of pre-exponential factors of reactions 2, (A_2) and reaction-1, (A_1), the same temperature with same steric factor (P) , $\left(\frac{A_2}{A_1}\right)^2$, is.

Given :

Species	Mass(g/mol)	Diameter (nm)
X	5	0.3
Y	20	0.5
M	10	0.4
N	10	0.4

Space for Rough Work

4. If K_{sp} of solution of AgCl is 1×10^{-12} . Number of solutions from the following which gives ppt. is.
- 0.1 M AgNO_3 (1 litre) + 0.1 M NaCl (10 litre)
 - 0.1 M AgNO_3 (10 litre) + 0.1 M NaCl (10 litre)
 - 10^{-4} M AgNO_3 (100 litre) + 10^{-4} M NaCl (100 litre)
 - 10^{-5} M AgNO_3 (10 litre) + 10^{-5} M NaCl (10 litre)
 - 10^{-6} M AgNO_3 (1 litre) + 10^{-6} M NaCl (1 litre)
 - 10^{-5} M AgNO_3 (2 litre) + 10^{-8} M NaCl (2 litre)
 - 10^{-5} M AgNO_3 (1 litre) + 10^{-7} M NaCl (4 litre)
 - 10^{-8} M AgNO_3 (2 litre) + 10^{-5} M NaCl (2 litre)
 - 10^{-4} M AgNO_3 (10 litre) + 10^{-4} M NaCl (10 litre)
5. Quinhydrone is a sparingly soluble one-to-one addition compound formed from hydroquinone and quinone. When solid quinhydrone is dissolved in an aqueous medium, equal concentrations of hydroquinone and quinone result. If a platinum wire is dipped into the solution, the potential of the electrode is govern by the reversible reaction :-

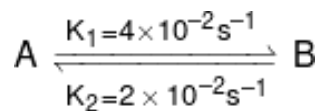


Where Q denotes quinone and H_2Q denotes hydroquinone. The E_{RP}^0 value of this half cell reaction is $+0.46 \text{ V}$ with respect to the saturated calomel reference electrode (SCE). Because the potential of the quinone-hydroquinone half reaction depends on the concentration of hydrogen ion, it is possible to use this system to measure pH. A sample solution of unknown pH was saturated with quinhydrone and a platinum electrode was dipped into it. If the potential of the such electrode was found to be $+0.22 \text{ V}$ with respect to SCE, what was the pH of the sample solution ?

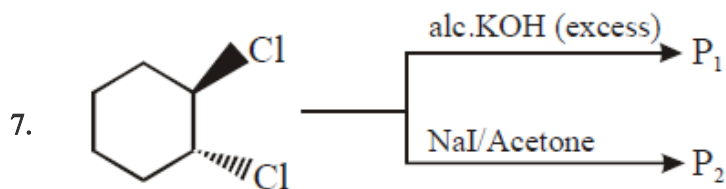
Given : $\frac{2.303RT}{F} = 0.06\text{V}$

Space for Rough Work

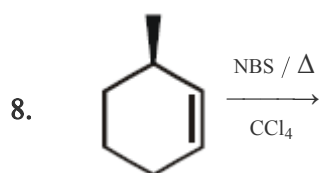
6. For a reversible reaction



if initial concentration of A is 21 mol L^{-1} . Then equilibrium concentration of A (in mol L^{-1}) will be.



X & Y is no. of double bond equivalent in P_1 & P_2 respectively, then $(X + Y)$ is.



Total mono brominated organic products obtained are ?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f(x, y) = \frac{1}{\lim_{\alpha \rightarrow 0} \left(\frac{\sqrt[3]{\sec \alpha} - \sqrt{\sec \alpha}}{\alpha^2} \right)}$, then

- (A) $f(2, 3) = 12$
- (B) $f(x, y) = 12$ has infinitely many solutions
- (C) Slope of tangent to the curve $f(x, y) = 12$ at $x = 2$ is 4
- (D) $f(x, y) = 12$ has no horizontal tangent

Space for Rough Work

2. If $\int \cot \left(2 \tan^{-1} \sqrt{\frac{\sqrt{1+\sqrt{x}} - x^{1/4}}{\sqrt{1+\sqrt{x}} + x^{1/4}}} \right) dx = \frac{qx^{p/q}}{p} + C$; (where p and q are relatively prime and C is constant of integration), then
- (A) $p = 5$ (B) $p + 2q = 13$ (C) $p - q = 1$ (D) $q = 5$
3. If $y^x - x^y = 1$, then $\frac{dy}{dx}$ at $x = 1$, is -
- (A) $2(1 - \ln 2)$ (B) $2(1 + \ln 2)$ (C) $\ln(4e^2)$ (D) $\ln\left(\frac{e^2}{4}\right)$
4. Identify the **CORRECT** statements -
- (A) Sum of two periodic functions may or maynot be a periodic function
- (B) A differentiable function in $\mathbb{R}$ will always be continuous in $\mathbb{R}$
- (C) A continuous function whose domain is in a closed interval is always bounded
- (D) In $[a, b]$ for a function $f(x)$, if $f(a) \cdot f(b) < 0$, then there exists atleast one root of $f(x) = 0$ in (a, b)

Space for Rough Work

5. If $\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 + \sqrt[3]{x})} dx = A \cdot x^B + C \tan^{-1}(x^D) + E$, then which of the following is/are true (where 'E' is constant of integration) ?
- (A) $A = \frac{2}{3}$
- (B) $AB = 1$
- (C) $(AB + C)$ is a prime number
- (D) Sum of all coefficients of even power of x in the expansion of $\left(AB + \frac{x}{D}\right)^C$ is equal to $\frac{(C+1)^C + (C-1)^C}{2}$
6. If $n \in \mathbb{N}$ and $f(x) = ({}^nC_0x + 1)({}^nC_1x + 2)({}^nC_2x + 3)({}^nC_3x + 4) \dots ({}^nC_nx + n + 1)$ and $f'(0) = \frac{3}{2}(7!)$, then n is equal to
- (A) 3 (B) 4 (C) 5 (D) 6

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $y = f(x)$ be a fourth degree algebraic polynomial function such that

(i) $\lim_{x \rightarrow 1} \frac{f(x)}{(x-1)^2} = 2$

(ii) It cuts y-axis a point where slope of normal is $\frac{1}{8}$

(iii) Tangent at $x = 2$ is parallel to $y = 8x + 1$

Consider $y = g(x)$ be a quadratic curve touching and cutting $y = f(x)$ on x-axis and y-axis respectively.

7. The angle of intersection between $y = f(x)$ & $y = g(x)$ at point where they cut y-axis is -

(A) $\tan^{-1} \frac{2}{33}$ (B) $\tan^{-1} \frac{2}{49}$ (C) $\tan^{-1} \frac{14}{33}$ (D) $\tan^{-1} \frac{14}{49}$

8. The abscissae of point on curve $y = g(x)$, where length of subtangent is 2018 can be -

(A) 1010 (B) 2019 (C) 3028 (D) 4037

Space for Rough Work

Paragraph for Questions 9 and 10

Let $f(x)$ be a polynomial function satisfying the following conditions $\lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0$, $\lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = -1$ and $f(0) = 0$ (Note : $[k]$ denotes greatest integer function less than or equal to k)

On the basis of above information, answer the following questions :

9. The number of points of discontinuity of $g(x) = [f(x)]$ in $[0, 3]$ is

(A) 2 (B) 3

(C) 4 (D) 5

10. The number of points of non-derivability of $h(x) = |f(|x|)|$ is -

(A) 2 (B) 3

(C) 4 (D) 5

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

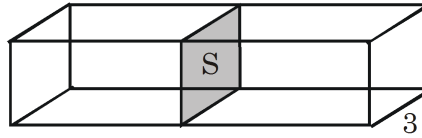
1. Let $J = \int \frac{x+y}{xy} dy$ and $I = \int \frac{x+y}{x^2} dx$ where x and y are independent variables such that $g\left(\frac{y}{x}\right) = J - I$ and $g(1) = 1$, then value of $g(e)$ is
2. Let $f(x)$ be a differentiable function on $[0, 8]$ such that $f(1) = 5$, $f(2) = \frac{1}{2}$, $f(3) = 6$, $f(4) = -3$, $f(5) = 4$, $f(6) = \frac{1}{4}$ and $f(7) = \frac{-1}{5}$. If the minimum number of roots of the equation $f'(x) - f'(x)(f(x))^2 = 0$ is λ , then $\frac{\lambda}{3}$ equals

Space for Rough Work

3. Let $f(x)$ be a real valued function satisfying $2f(xy) = (f(x))^y + (f(y))^x \forall x, y \in \mathbb{R}$, $f(1) = 3$ and $\lim_{x \rightarrow 0} \frac{f(x) + f(-x) - f(f(0)) + 1}{x^2}$ is equal to $(\ln k)^2$ where k is natural number, then the value of $\frac{k}{2}$ is
4. $\lim_{x \rightarrow \infty} x^3 \left(\sqrt{x^2 + \sqrt{1 + x^4}} - \sqrt{2x} \right) = \frac{\sqrt{2}}{\lambda}$ find value of λ
5. Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function differentiable at 3 and satisfying $f(3) = 3f'(3)$. If the value of $\lim_{x \rightarrow \infty} \left(\frac{f\left(3 + \frac{3}{x}\right)}{f(3)} \right)^x$ is L then $\frac{[L]}{5}$ is (where $[.]$ is greatest integer function)

Space for Rough Work

6. Let $f(x) = (x-1)^{100}(x-2)^{2(99)}(x-3)^{3(98)} \dots (x-100)^{100}$. If $P = \frac{f'(101)}{f(101)}$, then $\frac{P}{5000}$ equals
7. If a rectangular tank has a sliding panel S that divides it into two adjustable tanks of width 3 ft. See figure. Water is poured into the left compartment at the rate of $5 \text{ ft}^3/\text{min}$. At the same time S is moved to the right at the rate of $3 \text{ ft}/\text{min}$. When the left compartment is 10 ft long it contains 70 ft^3 of water, then how fast the water level is falling



8. If $f(x) + f\left(1 - \frac{1}{x}\right) = 1 + x \forall x \in \mathbb{R} - \{0, 1\}$, then value of $f(2)$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work